[redacted] Data Analysis in Python

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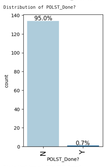
This project is a data analysis and machine learning model developed to generate valuable insights for key business functions. Prior to conducting any analysis, it is essential to thoroughly understand and prepare the data for processing and modeling. Leveraging Jupyter Notebook in Python ensures smooth and clear steps to ensure that the data is well-structured, consistent, and prepared for analysis or modeling.

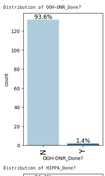
Importing libraries from Jupyter Notebook into python or PyCharm.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import sklearn
import sklearn.preprocessing
import sklearn.model_selection
import sklearn.metrics
import sklearn.ensemble
import sklearn.tree
import sklearn.linear_model
import sklearn.neighbors
import sklearn.svm
import sklearn.cluster
import sklearn.decomposition
import sklearn.feature_selection
import sklearn.pipeline
import sklearn.metrics
import sklearn.model_selection
import sklearn.preprocessing
```

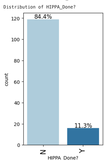
Distribution of POLST_Done?



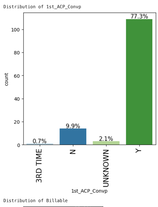
Distribution of OOH-DNR_Done?



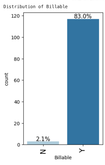
Distribution of HIPAA_Done?



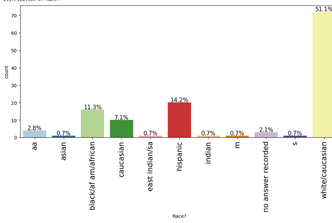
Distribution of 1st_ACP_Convp



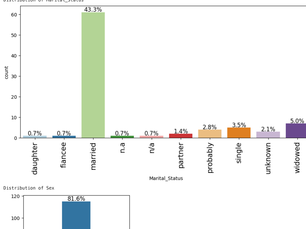
Distribution of Billable



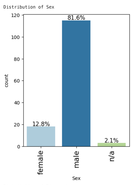
Distribution of Race?



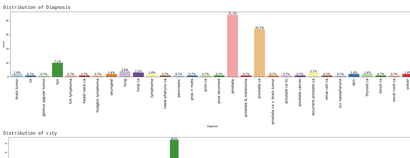
Distribution of Marital_Status



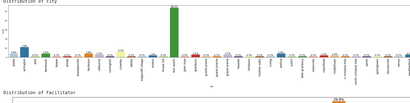
Distribution of Sex



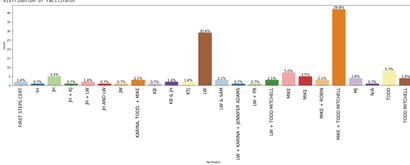
Distribution of Diagnosis



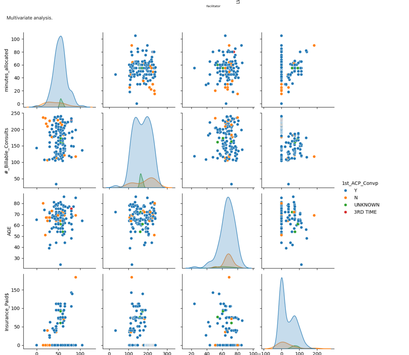
Distribution of city



Distribution of Facilitator



Multivariate analysis.



Summary Insights:

- **Minutes Allocated vs. #_Billable_Consults:** There seems to be no clear linear relationship between these two variables; points are spread out without any obvious trend.
- **#_Billable_Consults vs. Insurance_Paid$:** There is a slight pattern suggesting that as the number of billable consults increases, so does the insurance paid, although this is not strongly linear and has considerable scatter.
- **AGE vs. Other Variables:** Age does not show strong correlations with any other variables, indicating that age might be independent of other factors in this dataset.

Future Suggestions:

As per my conversations with Leah, below are some affordable and efficient methods that I think of, for moving client information from paperwork and Excel sheets to a more digital, centralized storage system:

1. Google Sheets with Google Forms and Google Drive

- **Cost**: Free with a Google account; $6/month per user for Google Workspace Basic.
- **How It helps**: Google Sheets can act as a more structured, shareable database than Excel. By using **Google Forms**, one can directly gather customer data digitally, avoiding paperwork altogether. Data from forms can be automatically saved to Google Sheets, which is easy to share and edit.
- **Additional Benefits**: With **Google Drive**, it can organize customer files (documents, agreements, etc.) in shared folders, making everything accessible in one place.

2. Microsoft Access (If Using Microsoft 365)

- **Cost**: Included with Microsoft 365 Business Standard and Business Premium plans.
- **How It helps**: Microsoft Access provides a straightforward database management system that integrates well with Excel. You can import Excel data and create forms for entering new data, queries for data filtering, and reports for analysis.
- **Additional Benefits**: Good for companies already using Microsoft 365, as Access integrates with other Microsoft tools like Outlook and Teams.

3. Zoho CRM

- **Cost**: Free plan available with limited features; paid plans start at $14/month per user.
- **How It helps**: Zoho CRM offers a dedicated system for storing and managing customer information, including contact details, sales history, and notes. It's a full-fledged CRM system that's affordable and can scale with the business as it grows.
- **Additional Benefits**: Zoho CRM includes lead management, email integration, and reporting features that can improve customer relationship management beyond simple data storage.

4. Convert Paper Records with Scanning + OCR Software

- **Cost**: One-time investment for a document scanner and OCR software (around 100- 200).
- **How It helps**: For companies with a lot of historical paperwork, using a scanner with OCR (Optical Character Recognition) software can digitize physical documents, making them searchable. You can then upload these documents to a cloud storage service (like Google Drive or Dropbox) or a database for easy access.
- **Additional Benefits**: OCR software turns scanned documents into searchable PDFs or editable text, making it easy to extract customer data and enter it into a digital system.

Summary:

Overall, all the digitilization methods mentioned above require ample amount of time and expertise to setup and streamline into Business As Usual. I believe, having a dedicated team will fasten the transition process. Before loading the raw data into any of the cloud services, it has to be cleaned and preprocessed, similar to what is done above, by using a python code.